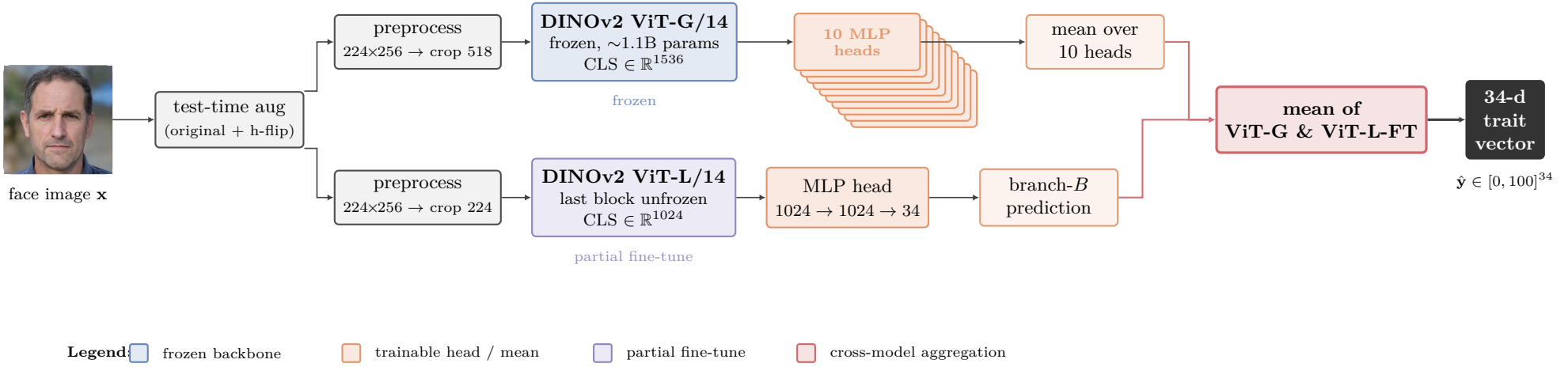
face-trait-transformer: image → 34-d perceived-trait vector

Cross-model ensemble of frozen ViT-G/14 heads + partially fine-tuned ViT-L/14, with horizontal-flip TTA



Training

- 803 train / 100 val stimuli (seed-0, 80/10/10)
- AdamW, cosine LR, batch 64, ≤400 epochs
- Loss: variance-weighted MSE, $w_{ij} = \mathrm{med}(\sigma)/(\sigma_{ij} + \varepsilon)$
- Early stop on val mean Pearson r (patience 40)

Sweep & selection

- 62 head configs / backbone (hidden × drop × lr × wd × seed)
- Top-10 chosen by *validation* mean r
- ViT-L fine-tune: head_lr = 10× backbone_lr
- Unweighted averaging ($\Delta_{\text{val-wt}} < 10^{-4}$)

Test performance (n = 101)

ViT-B baseline	0.757
ViT-L ensemble (10)	0.821
ViT-G-224 ensemble (10)	0.836
ViT-L fine-tune	0.835
ViT-G-518 (var-wt, 10)	0.848
Final (w/ TTA)	0.858